



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

FULL SYLLABUS

01-04-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE : [SCORE-II (ALL STAR BATCH)]

ANSWER KEY

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	B	D	D	B				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	B,C	A,C	A,C	A,C,D	B,C	B,C,D		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	9	3	8	3	4	1	6	9

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4				
	A.	B	C	B	D				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	A	A,D	A,C	A,B,D	A,B	A,D		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	0	6	5	8	9	9	6	3

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	C	B	B	A				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	B,D	A,B,C	A,C	A,B,C	A,C	B,D		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	9	7	7	1	1	5	7	2

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. **Ans (B)**

$$P_{\text{at top}} = P_{\text{at bottom}} + (\rho_{\text{perfume}}) g(\Delta h)$$

$$P_{\text{gauge}} = P_{\text{at top}} - P_{\text{at bottom}} + (\rho_{\text{perfume}}) g(\Delta h)$$

$$P_{\text{gauge}} = (\rho_{\text{perfume}}) g(\Delta h)$$

For the moving air, assume the location 1 is at the top of the nozzle and location 2 is far enough away for $v_2 = 0$.

$$P_1 + \frac{1}{2} \rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2} \rho v_2^2 + \rho g h_2$$

$$P_1 + \frac{1}{2} \rho_{\text{air}} v_1^2 = P_2$$

$$\frac{1}{2} \rho_{\text{air}} v_1^2 = P_2 - P_1$$

$$\rho_{\text{air}} v_1^2 = 2 (P_2 - P_1)$$

$$v_1^2 = \frac{2 (P_2 - P_1)}{\rho_{\text{air}}}$$

$$v_1^2 = \frac{2 (P_2 - P_1)}{\rho_{\text{air}}}$$

$$v_1 = \sqrt{\frac{2 (P_2 - P_1)}{\rho_{\text{air}}}}$$

$$v_1 = \sqrt{\frac{2 (\rho_{\text{perfume}} g h)}{\rho_{\text{air}}}}$$

$$= \sqrt{\frac{2 \times 910}{1.3} \times 10 \times \frac{3.5}{100}} = 7\sqrt{10}$$

2. Ans (D)

(a) The oscillations take place about the horizontal line through the point of suspension and perpendicular to the plane of the figure. The moment of inertia of the rod about this line is

$$\frac{m\ell^2}{12} + m\ell^2 = \frac{13}{12}m\ell^2.$$

$$\text{The time period} = 2\pi\sqrt{\frac{I}{mg\ell}} = 2\pi\sqrt{\frac{13m\ell^2}{12mg\ell}}$$

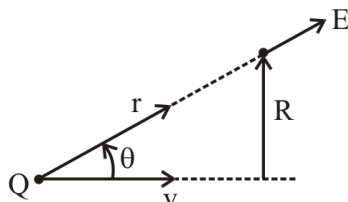
$$= 2\pi\sqrt{\frac{13\ell}{12g}}$$

(b) The angular oscillations take place about the suspension wire. The moment of inertia about this line is $\frac{m\ell^2}{12}$. The time period is

$$2\pi\sqrt{\frac{I}{k}} = 2\pi\sqrt{\frac{m\ell^2}{12k}}$$

$$\Rightarrow k = 13 mg\ell$$

3. Ans (D)



$$E = \frac{q}{4\pi\epsilon_0 r^2}$$

$$B = \frac{\mu_0 q v r \sin \theta}{4\pi r^3}$$

$$\vec{S} = \frac{1}{\mu_0} (\vec{E} \times \vec{B}) = \frac{q^2 v R}{16\pi^2 \epsilon_0 r^5}$$

$$S_{\max} = \frac{q^2 v}{16\pi^2 \epsilon_0 R^4}$$

$$= \frac{81 \times 10^{-12} \times \pi \times 10^{-3} \times 9 \times 10^9}{4\pi \times 81 \times 10^{-8}}$$

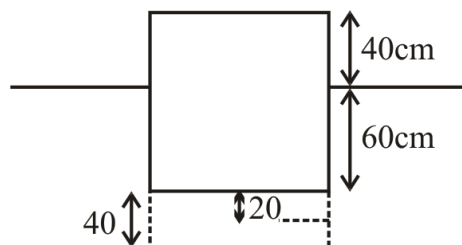
$$\frac{9}{4} \times 10^2 = 225 \text{ W/m}^2$$

4. Ans (B)

$$mg + \rho v g = \rho_0 V_{LD} g$$

$$m = 1000 \times 0.60 - 500 \times 1 = 100 \text{ kg.}$$

$$mv = \rho V \times v'$$



$$\omega_B = \omega_{gr} \text{ on liquid}$$

$$= -\rho_0 \times 1 \times 0.80 \times g \times 0.40$$

$$= \Delta k = -\frac{1}{2} \times \rho \times 1^3 \times v'^2$$

$$\Rightarrow v' = \sqrt{20 \times 0.80 \times 0.40 - \frac{500g \times 0.4}{500}}$$

$$v = \sqrt{2gh}$$

$$h = 6\text{m}$$

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (B,C)

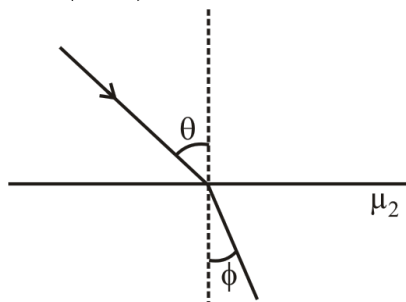
$$i_L = \frac{V}{R} \left(1 - e^{-\frac{Rt}{L}} \right) \quad t < \frac{L}{R} \ln 2$$

$$\Rightarrow \epsilon = v^{-\frac{Rt}{L}} = \frac{V}{4}$$

$$i_L = \frac{V}{2R}$$

$$i' = \frac{V}{2R} e^{-\frac{3Rt}{L}} = \frac{V}{16R}$$

6. Ans (A,C)



$$\hat{e}_i = \frac{3\hat{i} - 4\hat{j}}{5} \Rightarrow \tan \theta = \frac{3}{4}$$

$$\theta = 37^\circ$$

$$\mu_1 = \frac{C}{u_1} = 2 = \sqrt{\epsilon_r}$$

$$\mu_1 \tan 37^\circ = \mu_2$$

$$2 \times \frac{3}{4} = \mu_2$$

$$\Rightarrow \mu_2 = 1.5 = \sqrt{\epsilon_r}$$

$$2 \sin \theta = 1.5 \sin \phi$$

$$\Rightarrow 2 \times \frac{3}{5} = 1.5 \sin \phi$$

$$\Rightarrow \phi = 53^\circ$$

$$\vec{E}_i \text{ is } \perp \text{ to } \vec{k} \Rightarrow \hat{E}_i = \frac{4}{5}\hat{i} + \frac{3}{5}\hat{j} \text{ \& } \hat{E}_r = \frac{3}{5}\hat{i} + \frac{4}{5}\hat{j}$$

$$\text{no reflected ray} \Rightarrow I_i = I_r$$

$$\frac{1}{2} u_1 \epsilon_0 \epsilon_r E_i^2 = \frac{1}{2} u_2 \epsilon_0 \epsilon_r E_r^2$$

$$\Rightarrow 1.5 \times 10^8 \times 2^2 \times 3^2 = 2 \times 10^8 \times 1.5^2 \times E_r^2$$

$$54 = 9 E_r^2$$

$$E_r = \sqrt{6}$$

7. Ans (A,C)

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \frac{2\pi}{\lambda} \left(\frac{dy}{D} - (\mu - 1)t \right)$$

$$= 0.25 + 0.64 \times 0.25 + 2\sqrt{0.25^2 \times 0.64}$$

$$\cos \left(\frac{2\pi}{5 \times 10^{-7}} \right) \left(\frac{10^{-3}y}{1} - 5 \times 10^{-6} \right)$$

$$= 0.41 + 0.4 \cos (2\pi (2y - 10)) \text{ (y in mm)}$$

8. Ans (A,C,D)

$$\frac{1}{80} + \frac{1}{20} = \frac{1}{8} = \frac{1+4}{80} \Rightarrow f = 16 \text{ cm}$$

$$\frac{1}{16} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\frac{4}{-3 \times 40} - \frac{1}{-80} = \frac{1-\mu}{R_2} + \frac{\mu - \frac{4}{3}}{R_1} \dots (i)$$

$$\frac{-1}{48} = \frac{-1}{30} + \frac{1}{80} = \frac{1-\mu}{R_2} + \frac{\mu - \frac{4}{3}}{R_1} \dots (ii)$$

$$\frac{7}{80} + \frac{4}{3 + -80} = \frac{\frac{4}{3} - \mu}{R_2} + \frac{\mu - 1}{R_1}$$

$$\frac{5}{48} = \frac{21+4}{240} = \frac{\frac{4}{3} - \mu}{R_2} + \frac{\mu - 1}{R_1} \dots (iii)$$

$$\frac{1}{24} = \frac{\frac{4}{3} - \mu}{R_2} + \frac{(\mu - 1)}{R_2} = \frac{1}{3R_2} \Rightarrow R_2 = 8 \text{ cm}$$

$$\frac{-1}{16} - \frac{1}{48} = \frac{\mu - \frac{4}{3}}{R_1} - \frac{(\mu - 1)}{R_1} = \frac{-1}{3R_1}$$

$$+\frac{1}{12} = +\frac{1}{3R_1}$$

$$R_1 = 4 \text{ cm}$$

$$\frac{1}{16} = (\mu - 1) \left(\frac{1}{4} - \frac{1}{8} \right)$$

$$\frac{1}{16} = (\mu - 1) \left(\frac{2}{8} - \frac{1}{8} \right)$$

$$\mu = 1.5$$

9. Ans (B,C)

$$k_{\max} = \frac{hC}{\lambda} - \phi = 4 - 1.5 = 2.5 \text{ eV}$$

$$eEd = k_{\max} = 2.5 \text{ eV}$$

$$\frac{\sigma}{\epsilon_0} \times d = 2.5$$

$$d = 0.25 \text{ m}$$

$$h_{\max} = \frac{v^2 \sin^2 \theta}{2g} = \frac{mv^2}{2eE} \sin^2 \theta$$

$$h_{\max} = \frac{k}{eE} \sin^2 \theta = \frac{2.5}{10} \times \frac{3}{4} = \frac{3}{16} \text{ m}$$

PART-1 : PHYSICS

SECTION-III

1. Ans (9)

$$h = \frac{25 \cos \theta}{rpg} \Rightarrow \frac{\Delta S}{S} = \frac{\Delta h}{h} + \frac{\Delta r}{r}$$

$$= \frac{2 \times 0.01}{2.00} \times 100 + \frac{2 \times 0.01}{0.25} \times 100$$

2. Ans (3)

$$\begin{aligned}\vec{v}_w - \vec{v}_b &= 4\hat{i} \\ \vec{v}_w &= 4\hat{i} + \vec{v}_b \\ &= 4\hat{i} + 5 \left[-\frac{4}{5}\hat{i} + \frac{3}{5}\hat{j} \right] = 3\hat{j}\end{aligned}$$

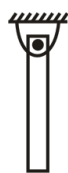
3. Ans (8)

$$\begin{aligned}v &= \frac{\omega}{R} = 50 = \sqrt{\frac{TL}{m}} \\ L &= \frac{2500 \times 30 \times 10^{-3}}{10} = 7.5\text{m} \\ \lambda &= \frac{2\pi}{R} = 2\text{m} \\ LL &= \frac{\lambda}{2} = 1\text{m} \\ &\Rightarrow 7\frac{1}{2}\text{ loops} \\ &\Rightarrow 8\text{ antinodes.}\end{aligned}$$

4. Ans (3)

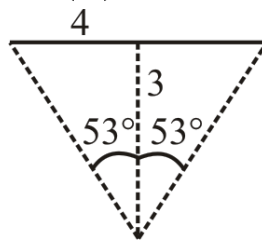
$$\begin{aligned}\frac{dU}{dx} &= \frac{-a}{x^2} + \frac{2b}{x^3} = 0 \Rightarrow x = \frac{2b}{a} \\ U_0 &= \frac{b \times a^2}{4b^2} - \frac{a^2}{2b} = -\frac{a^2}{4b} \\ U_0 + K_0 &= U_\infty + K_\infty \\ \frac{1}{2}mv^2 &= \frac{a^2}{4b} \\ v &= \sqrt{\frac{a^2}{2bm}} = \frac{a}{\sqrt{2bm}} = \frac{6}{\sqrt{2 \times 2 \times 1}} = 3\text{m/s}\end{aligned}$$

5. Ans (4)



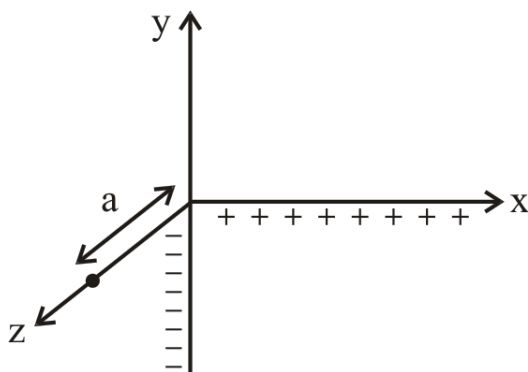
$$\begin{aligned}F_i &= mg \\ F_f - mg &= \frac{m\omega^2 \ell}{2} \\ 2g &= \frac{\omega^2 \ell}{2} \\ \omega &= \sqrt{\frac{4g}{\ell}} = 10\text{rad/s} \\ J\ell &= I\omega = \frac{m\ell^2}{3}\omega \\ J &= 3 \times \frac{0.4}{3} \times 10 = 4\text{kg m/s}\end{aligned}$$

6. Ans (1)



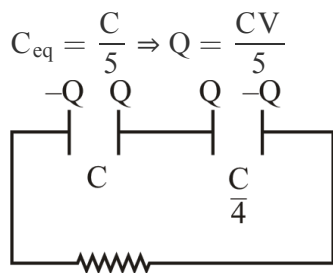
$$\begin{aligned}B &= \frac{\mu_0 i}{4\pi \times 3} [\sin 53 \times 2] \hat{i} \\ &+ \frac{\mu_0 i}{4\pi \times 3} [\sin 53 \times 2] \hat{j} \\ &+ \frac{\mu_0 i}{4\pi \times 4} [\sin 37 + \sin 0] \hat{i} \times 2 \\ &+ \frac{\mu_0 i}{4\pi \times 4} [\sin 37 + \sin 0] \hat{i} \times 2 \\ &= 10^{-7} \times i \left[\frac{8}{15} \hat{i} + \frac{8}{15} \hat{j} + \frac{3}{10} \hat{i} + \frac{3}{10} \hat{j} \right] \\ &= 10^{-7} \times i \left[\frac{25}{30} (\hat{i} + \hat{j}) \right] \\ B &= 10^{-7} \times \frac{5}{3\sqrt{2}} = \frac{30\sqrt{2}}{3\sqrt{2}} \times 10^{-7} = 1\mu\text{T}\end{aligned}$$

7. Ans (6)



$$\begin{aligned}\vec{E} &= \frac{\lambda}{4\pi\epsilon_0 a} [\hat{k} - \hat{i}] \\ &+ \frac{2\lambda}{4\pi\epsilon_0 a} [-\hat{k} - \hat{j}] \\ &= \frac{\lambda}{4\pi\epsilon_0 a} [-\hat{k} - \hat{i} - 2\hat{j}] \\ E &= \frac{\sqrt{6}\lambda}{4\pi\epsilon_0 a} \\ n &= 6\end{aligned}$$

8. Ans (9)



In steady state they are in parallel.

$$\Rightarrow 2Q = \left(C + \frac{C}{4}\right) V'$$

$$\Rightarrow 2 \times \frac{CV}{5} = \frac{5C}{4} V'$$

$$V' = \frac{8V}{25}$$

$$U_f = \frac{1}{2} \times \frac{5C}{4} \times V'^2 = \frac{5C}{8} \times \frac{64V^2}{625} = \frac{8CV^2}{125}$$

$$U_i = \frac{1}{2} \times \frac{C}{5} \times V^2 = \frac{CV^2}{10}$$

$$\Delta H = -\Delta U = CV^2 \left(\frac{1}{10} - \frac{8}{125} \right)$$

$$= CV^2 \left(\frac{25 - 16}{250} \right) = \frac{9CV^2}{250}$$

$$= \frac{9 \times 0.1 \times 2500}{250} = 9 \mu J$$

PART-2 : CHEMISTRY

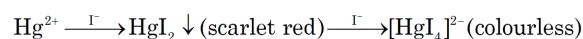
SECTION-I (i)

1. Ans (B)

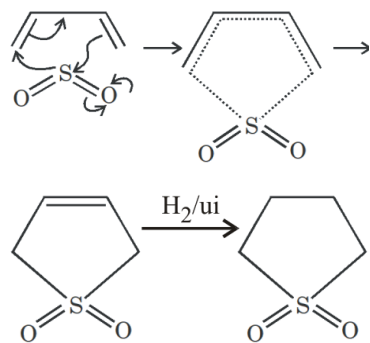
$$\Delta G^\circ_{\text{cell}} = -RT \ln K_{\text{sp}} = -nF E^\circ_{\text{cell}}$$

$$E^\circ_{\text{cell}} = \frac{+2.303RT}{3F} \log K_{\text{sp}} = \frac{0.06}{3} (-36.4) = -0.728 \text{ V}$$

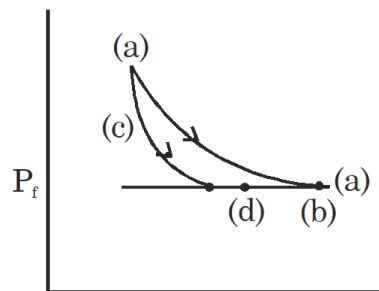
2. Ans (C)



3. Ans (B)



4. Ans (D)



The graph indicates final states when system undergoes given processes.

So, final temperature are in order

$$T_a = T_b > T_d > T_c$$

PART-2 : CHEMISTRY

SECTION-I (ii)

5. Ans (A)

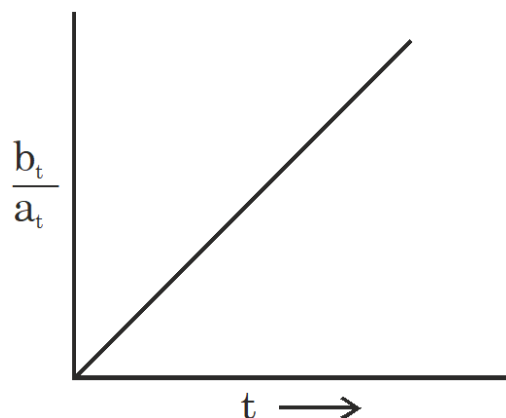
It is E_2 elimination

We know from 2nd order kinetics that,

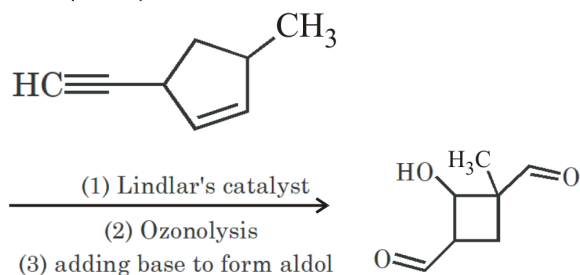
$$\left(\frac{1}{a_t} - \frac{1}{a_0} \right) = kt$$

$$\Rightarrow \frac{a_0 - a_t}{a_t} = ka_0 \cdot t$$

$$\Rightarrow \frac{b_t}{a_t} = ka_0 \cdot t \quad \dots (1)$$



6. Ans (A,D)



7. Ans (A,C)

S_1 contains $S_X : X_Y = 2 : 3$

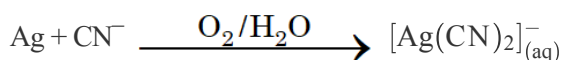
$$\therefore m(S_1) = \frac{\frac{5}{M_X}}{95 \times 10^{-3}} \times \frac{2}{5} \times \frac{\frac{5}{M_Y}}{95 \times 10^{-3}} \times \frac{3}{5}$$

$$\text{and } m(S_2) = \frac{\frac{5}{M_X}}{95 \times 10^{-3}} \times \frac{3}{5} \times \frac{\frac{5}{M_Y}}{95 \times 10^{-3}} \times \frac{2}{5}$$

$$\therefore \frac{m(S_1)}{m(S_2)} = \frac{\frac{10}{M_X} + \frac{15}{M_Y}}{\frac{15}{M_X} + \frac{10}{M_Y}} = \frac{2 + 3 \left(\frac{M_X}{M_Y} \right)}{3 + 2 \left(\frac{M_X}{M_Y} \right)}$$

$$= \frac{2 + 3 \left(\frac{1}{3} \right)}{3 + 2 \left(\frac{1}{3} \right)} = \frac{3}{\frac{11}{3}} = \frac{9}{11}$$

8. Ans (A,B,D)



Reduction potential of H_2O is greater than Ca^{+2} thus Ca^{+2} cannot be reduced in its aqueous solution.

9. Ans (A,B)

$$[1-x] = \left[\frac{Xxk_1}{k_1 + k_2} \right] \therefore [A]_t = [B]_t$$

At time t ,

$$= [1-x] = \frac{xk_1}{k_1 + k_2}$$

$$\Rightarrow 0.4 = \frac{0.6k}{k_1 + k_2} \Rightarrow \frac{k_1}{k_2} = 0.5$$

also $r = k[A]'$

$$\Rightarrow \frac{dx}{dt} = (k_1 + k_2)[A]'$$

$$\text{At time, } \Rightarrow \frac{0.8}{10} = (k_1 + k_2)[0.4]$$

$$k_1 + k_2 = 0.2$$

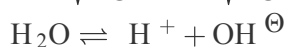
$$\Rightarrow 0.5k_2 + k_2 = 0.2$$

$$\Rightarrow 1.5k_2 = 0.2 \Rightarrow k_2 = \frac{0.2}{1.5}$$

$$t_{1/2} = \frac{0.693}{k_1 + k_2} = \frac{0.693}{0.2} = 3.5 \text{ mm}$$

10. Ans (A,D)

$$\alpha = \sqrt{\frac{K_a}{C}}, \alpha = \sqrt{\frac{K_b}{C}}$$



(H^+) is less in 1MHCN than 1MHCL

$$\alpha = \sqrt{\frac{K_a}{C}}$$

$T \uparrow \text{ at}$

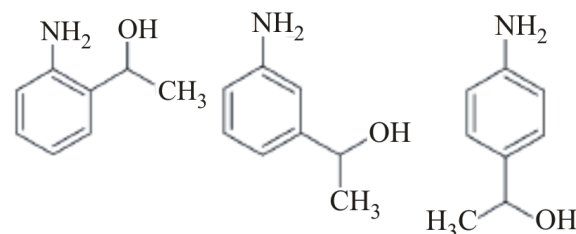
PART-2 : CHEMISTRY

SECTION-III

1. Ans (0)

Cupric ion forms brown precipitate

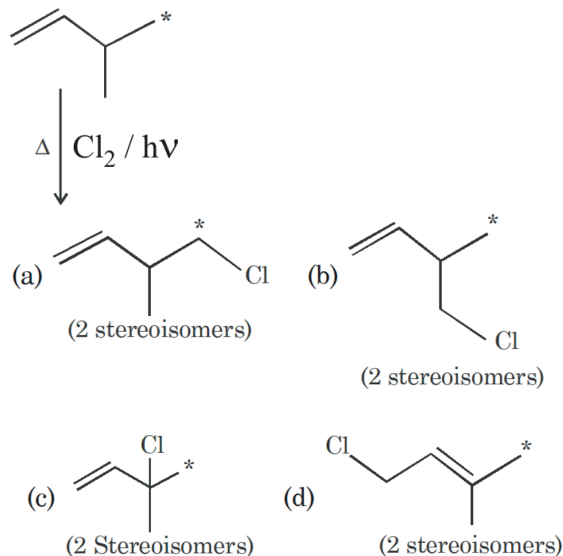
2. Ans (6)



3. Ans (5)

$$\Delta G^\circ = -25660 + 20 \times 1283 = 0 \Rightarrow K_{eq} = 1$$

4. Ans (8)



No free radical attack on double bond. Total 8 products are formed

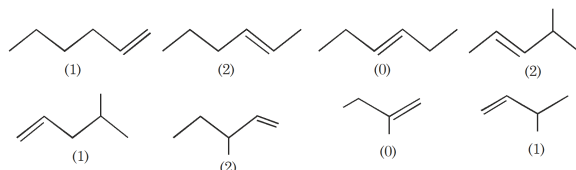
5. Ans (9)

$$V_m = 10b \Rightarrow b = \frac{V_m}{10}$$

$$\text{So, } P V_m \left(\frac{9}{10} \right) RT \Rightarrow \frac{P V_m}{RT} = \frac{10}{9}$$

$$\therefore Z = \frac{V_R}{V_i} = \frac{10}{9} \Rightarrow V_i = \frac{9V_R}{10} \Rightarrow V_i = 9b$$

6. Ans (9)



7. **Ans (6)**

$$EAN = 26 - 2 + (6 + 2 + 4) = 36 = X$$

8. **Ans (3)**

Only 12.09 eV photon can be used to excite electron.

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (C)**

$$y(xy - (x^2 - y^2)^2) \cdot dx = \{y^3 - x(x^2 - y^2)^2\} dy$$

$$\Rightarrow xy^2 dx - y(x^2 - y^2)^2 dx = y^3 dy - x(x^2 - y^2)^2 dy$$

$$\Rightarrow y^2 (x dx - y dy) = (x^2 - y^2)^2 (y dx - x dy)$$

$$\Rightarrow x dx - y dy = (x^2 - y^2)^2 \frac{(y dx - x dy)}{y^2}$$

$$\frac{2(x dx - y dy)}{(x^2 - y^2)^2} = \frac{2(y dx - x dy)}{y^2}$$

$$\int \frac{d(x^2 - y^2)}{(x^2 - y^2)^2} = 2 \int d\left(\frac{x}{y}\right)$$

$$\frac{-1}{x^2 - y^2} = \frac{2x}{y} - C$$

$$\therefore \frac{2x}{y} + \frac{1}{x^2 - y^2} = C$$

2. **Ans (B)**

$$\text{Let } \vec{r} = \alpha \vec{a} + \beta \vec{b} + \lambda(\vec{a} \times \vec{b})$$

$$\Rightarrow 4\alpha + \beta = 2$$

$$\text{also } \alpha + \beta = 8$$

$$\therefore \alpha = -2, \beta = 10$$

$$\text{also } \lambda = 2$$

$$\text{hence } \vec{r} = -2\vec{a} + 10\vec{b} + 2(\vec{a} \times \vec{b})$$

3. **Ans (B)**

$$\cot(3\alpha - 7\beta) \tan(5\beta - \alpha) = 1$$

$$\cos(3\alpha - 7\beta) \sin(5\beta - \alpha) = \sin(3\alpha - 7\beta) \cos(5\beta - \alpha)$$

$$= \sin[(5\beta - \alpha) - (3\alpha - 7\beta)] = 0$$

$$\Rightarrow \sin(4\alpha - 12\beta) = 0$$

$$\Rightarrow 4(\alpha - 3\beta) = n\pi$$

$$\Rightarrow \frac{\alpha - 3\beta}{3} = \frac{n\pi}{12}$$

$$\Rightarrow \alpha - 3\beta \in (-\pi, \pi)$$

$$\Rightarrow -4 < n < 4$$

$$\Rightarrow n = -3, -2, -1, 0, 1, 2, 3$$

$$\text{Now } 2\sqrt{2} \sin\left(\frac{n\pi}{12}\right) \text{ is integer for } n = 0, 3, -3$$

4. **Ans (A)**

$$\text{Given equation } \Rightarrow \tan^{-1}(a^2 + k) = \tan^{-1}|a| + \frac{\pi}{4}$$

Case (1): If $|a| = 0 \Rightarrow k = 1$

Case (2): If $|a| = 1 \Rightarrow k \in \phi$

$$\text{Case (3): If } |a| = 2 \Rightarrow \tan^{-1}|a| + \frac{\pi}{4} = \tan^{-1}(a^2 + k)$$

$$\Rightarrow \pi + \tan^{-1} \frac{|a| + 1}{1 - |a|} = \tan^{-1}(k + a^2)$$

$$\Rightarrow \pi - \tan^{-1} 3 = \tan^{-1}(k + 4); \text{ not possible as LHS } \in \left(\frac{\pi}{2}, \pi\right) \text{ and RHS}$$

$$\in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. **Ans (B,D)**

$$3ABA^{-1} + 3A = 2A^{-1}BA + 2A$$

$$3ABA^{-1} + 3A AA^{-1} = 2A^{-1}BA + 2A^{-1}AA$$

$$|3A(B+A)A^{-1}| = |2A^{-1}(B+A)A|$$

$$3^n|A+B| = 2^n|A+B|$$

$$\Rightarrow |A+B| = 0$$

$$(C) 3ABA^{-1} - 3A^{-1}BA = -A^{-1}BA - A$$

$$= -A^{-1}BA - A^{-1}AA$$

$$3^n|ABA^{-1} - A^{-1}BA| = (-1)^n|B+A| = 0$$

6. **Ans (A,B,C)**

$\therefore$ Domain of function is real number so $a = 0$ and

$$b^2 - 132 < 0$$

$$\text{Now } y = \frac{x^2 + 9bx + 17}{x^2 + bx + 33}$$

$$\Rightarrow (y-1)x^2 + (by-9b)x + (33y-17) = 0$$

$$\therefore x \in \mathbb{R} \text{ so } D \geq 0 \Rightarrow (by-9b)^2 - 4(y-1)(33y-17) \geq 0$$

$$\Rightarrow (132-b^2)y^2 + (18b^2-200)y - (81b^2-68) \leq 0$$

$$\text{and } y \in [-1, 2]$$

$$\text{Hence } b^2 = 4$$

$$\therefore b = 2 \text{ or } -2$$

$$\therefore \text{For } b = 2 \Rightarrow d = -5, e = 7$$

$$\text{For } b = -2 \Rightarrow d = -7, e = 5$$

7. Ans (A,C)

$$\frac{20}{1}, \frac{20}{2}, \frac{20}{3}, \frac{20}{4}, \frac{20}{5} : \text{H.P.}$$

$$0 + \frac{1}{2} + \frac{1}{4} + \left(\frac{5}{8} + \frac{5}{16} + \frac{5}{32} + \dots \right) = 2$$

$$H_2 = \frac{20}{3}; [H_2] = 6; \left[\frac{H_2}{2} \right] = 3; \left[\frac{H_2}{4} \right] = 1$$

8. Ans (A,B,C)

$$f(x) \cdot f(y) + f(x+y) = e^x f(y) + e^y f(x) + xy$$

$$\therefore f(x) \cdot f'(y) + f'(x+y) = e^x f'(y) + e^y f'(x) + x$$

Put $y = 0$

$$f(x) \cdot f'(0) + f'(x) = e^x f'(0) + f(x) + x$$

$$\Rightarrow f'(x) - f(x) = x \Rightarrow f(x) = e^x - x - 1$$

$$\therefore (A) \lim_{x \rightarrow 0} \frac{f(x)}{x^2} = \lim_{x \rightarrow 0} \frac{e^x - x - 1}{x^2} = \frac{1}{2}$$

$$(B) \int_x^{x^2} t dt = \left(\frac{t^2}{2} \right)_x^{x^2} = \frac{x^4 - x^2}{2} > 0 \forall |x| > 1$$

$$(C) F(x) = f'(x) - f(x) = (e^x - 1) - (e^x - x - 1) = x$$

which is increasing function so

$$F(x_2) > F(x_1) \text{ if } x_2 > x_1$$

$$(D) \text{ Now } f(x) = e^x - 1 \text{ and } f''(x) = e^x > 0 \forall x \in \mathbb{R}$$

$\therefore f(x)$ is increasing function on $\mathbb{R}$

$$\therefore f(-1) = -ve \text{ and } f(1) = +ve$$

$$\therefore f(c) = 0 \text{ has exactly one root in } (-1, 1)$$

i.e., one horizontal tangent in $(-1, 1)$

9. Ans (A,C)

$$\text{Let } z^2 + 4z = t$$

$$|2t + 1| = 1 \Rightarrow 0 \leq |t| \leq 1$$

$$\Rightarrow 0 \leq |z^2 + 4z| \leq 1$$

$$\Rightarrow 0 \leq |z||z + 4| \leq 1$$

$$\text{If } |z| = 5 \text{ then } |z + 4| \leq \frac{1}{5}, \text{ not possible}$$

$$\text{as } |z + 4| \geq ||z| - 4|$$

$$\text{If } |z| = 3 \text{ then } |z + 4| \leq \frac{1}{3}, \text{ not possible}$$

10. Ans (B,D)

$$(A) \text{ Not always true of } f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right) & x \neq 0 \\ 0 & x = 0 \end{cases}$$

$$(B) \text{ e.g. } f(x) = [x], g(x) = \{x\}, x = 0$$

$$(C) \lim_{x \rightarrow b} g(x) \text{ may not be equal to } \lim_{x \rightarrow b} g(x)$$

(D) $f(x) - g(x)$ needs to be continuous for a solution to always exist

PART-3 : MATHEMATICS

SECTION-III

1. Ans (9)

The equation of line L be $y - 2 = m(x - 8), m < 0$

co-ordinates of P and Q are

$$P \left(8 - \frac{2}{m}, 0 \right) \text{ and } Q(0, 2 - 8m).$$

$$\text{So, } OP + OQ = 8 - \frac{2}{m} + 2 - 8m$$

$$= 10 + \frac{2}{(-m)} + 8(-m) \geq 10 + 2\sqrt{\frac{2}{(-m)} \times 8(-m)} \geq 18$$

So, absolute minimum value of

$$OP + OQ = 18$$

2. Ans (7)

$$g(x_1)g(x_2)g(x_3)g(x_4)g(x_5)$$

$$= (x_1^2 - 2)(x_2^2 - 2)(x_3^2 - 2)(x_4^2 - 2)(x_5^2 - 2)$$

$$\text{Let } x^2 - 2 = y$$

$$\Rightarrow x = \sqrt{y + 2}, \text{ put in } p(x) = 0$$

$$(\sqrt{y + 2})^5 + y + 2 + 1 = 0$$

$$(y + 2)^5 = (y + 3)^2$$

$$\text{Product of roots} = -(32 - 9) = -23$$

$$\text{Now, } x_1 x_2 x_3 x_4 x_5 = -1 \Rightarrow g(x_1 x_2 \dots x_5) = -1$$

$$\Rightarrow g(x_1)g(x_2) \dots g(x_5) - 30g(x_1 x_2 \dots x_5) = 7$$

3. Ans (7)

$2x - y = 0$ is tangent passing through $(0, 0)$.

Line joining $(-2, 1)$ & $(0, 0)$ is perpendicular to

$$2x - y = 0$$

$\Rightarrow$ vertex is $(0, 0)$

4. **Ans (1)**

Let $1 - x^3 = t$

$$3I_1 = \int_0^1 (1 - t^{\sqrt{2}})^{\sqrt{3}} dt \text{ and } 3I_2 = \int_0^1 (1 - t^{\sqrt{2}})^{\sqrt{3}+1} dt$$

$$= - \int_0^1 (\sqrt{3}+1) \sqrt{2} (1 - t^{\sqrt{2}})^{\sqrt{3}} (1 - t^{\sqrt{2}} - 1) dt$$

(Using integration by parts)

$$3I_2 = -(\sqrt{3}+1) \sqrt{2} 3I_2 + (\sqrt{3}+1) \sqrt{2} 3I_1$$

$$\text{So, } \frac{I_1}{I_2} - \frac{\sqrt{3}-1}{2\sqrt{2}} = 1 + \frac{\sqrt{3}-1}{2\sqrt{2}} - \frac{\sqrt{3}-1}{2\sqrt{2}} = 1$$

5. **Ans (1)**

Each prime can go in one of the five slots = x_1 , x_2 , x_3 , x_4 and none of these hence total solution is equals to $5^4 = 625$

6. **Ans (5)**

Maximum exponent is 2047

So we wish to exclude x^{31}

excluding $x^1, x^2, x^4, x^8, x^{16}$ is only way

$$\text{Coefficient} = 1.2.4.8.16 = 1024$$

7. **Ans (7)**

2 blue result occur on or before 10 roll is equivalent to the occurrence of blue face at least twice in 10 rolls

$$n = 10, p = \frac{1}{3}, q = \frac{2}{3}$$

$$P(m \geq 2) = 1 - [P(0) + P(1)] = \frac{3^9 - 2^{11}}{3^9}$$

8. **Ans (2)**

$x - y + 2 = 0$ & $x - y - 2 = 0$ are both tangents

$\Rightarrow x - y = 0$ passes through centre & $3x + y = 4$

also passes through centre

$$\Rightarrow (g, f) \equiv (1, 1)$$